

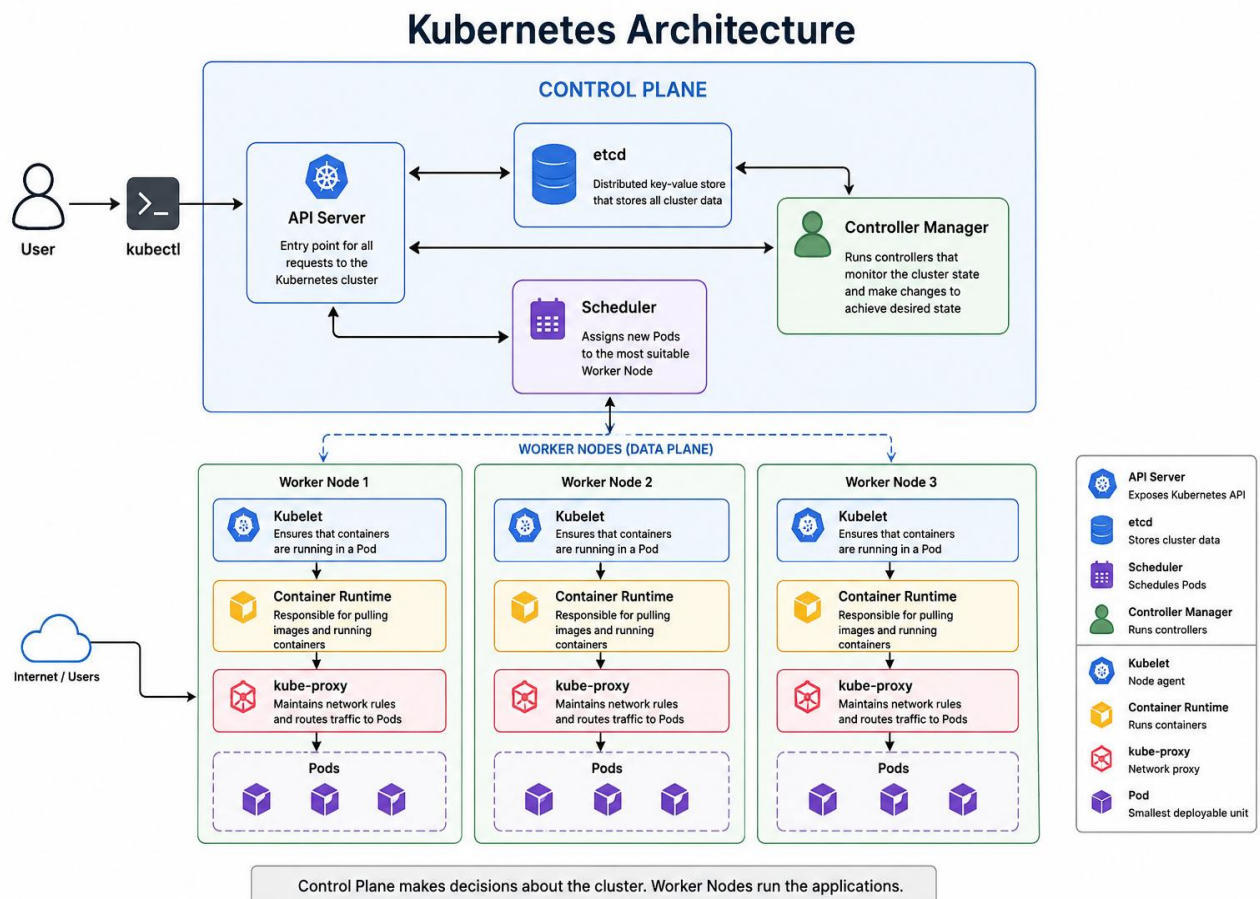
Kubernetes Architecture

Overview

A Kubernetes cluster consists of two major components:

1. Control Plane
2. Worker Nodes (Data Plane)

The Control Plane manages the cluster, while Worker Nodes run the actual applications.



Control Plane

The Control Plane is the brain of Kubernetes. It makes decisions about the cluster and maintains the desired state.

Components

- API Server
 - etcd
 - Scheduler
 - Controller Manager
-

API Server

The API Server is the entry point of Kubernetes.

Responsibilities

- Receives requests from users and components
 - Validates requests
 - Updates cluster state
 - Communicates with other control plane components
-

etcd

etcd is a distributed key-value store used by Kubernetes.

Responsibilities

- Stores cluster state
 - Stores configuration data
 - Stores resource definitions
 - Stores metadata
-

Scheduler

The Scheduler assigns Pods to Worker Nodes.

Responsibilities

- Selects the best node

- Considers CPU and memory availability
 - Applies scheduling policies
-

Controller Manager

The Controller Manager runs Kubernetes controllers.

Responsibilities

- Monitors resources
 - Maintains desired state
 - Detects failures
 - Performs corrective actions
-

Worker Node (Data Plane)

Worker Nodes are responsible for running application workloads.

Components

- Kubelet
 - Container Runtime
 - kube-proxy
 - Pods
-

Kubelet

Kubelet is the node agent that runs on every worker node.

Responsibilities

- Communicates with the API Server
 - Ensures Pods are running
 - Monitors Pod health
 - Reports node status
-

Container Runtime

The Container Runtime is responsible for running containers.

Examples

- containerd
- CRI-O

Responsibilities

- Pull images
- Create containers
- Start and stop containers

Interview Question

kube-proxy

kube-proxy is the networking component of Kubernetes.

Responsibilities

- Maintains networking rules
 - Routes traffic to Pods
 - Enables Service communication
-

Pod

A Pod is the smallest deployable unit in Kubernetes.

Features

- Contains one or more containers
- Shares network namespace
- Shares storage
- Shares lifecycle

Interview Question

Kubernetes Request Flow

When a user executes:

```
kubectl apply -f deployment.yaml
```

The request follows this flow:

```
kubectl
↓
API Server
↓
etcd
↓
Scheduler
↓
Worker Node
↓
Kubelet
↓
Container Runtime
↓
Pod Running
```

Architecture Summary

Control Plane

- API Server
- etcd
- Scheduler
- Controller Manager

Worker Node

- Kubelet
- Container Runtime
- kube-proxy
- Pods

Key Point

Control Plane makes decisions.

Worker Nodes execute those decisions.